

20250930 From SF to HII region

Note that the y-axis range in each panel is no longer identical here so that we can zoom in the see more details.

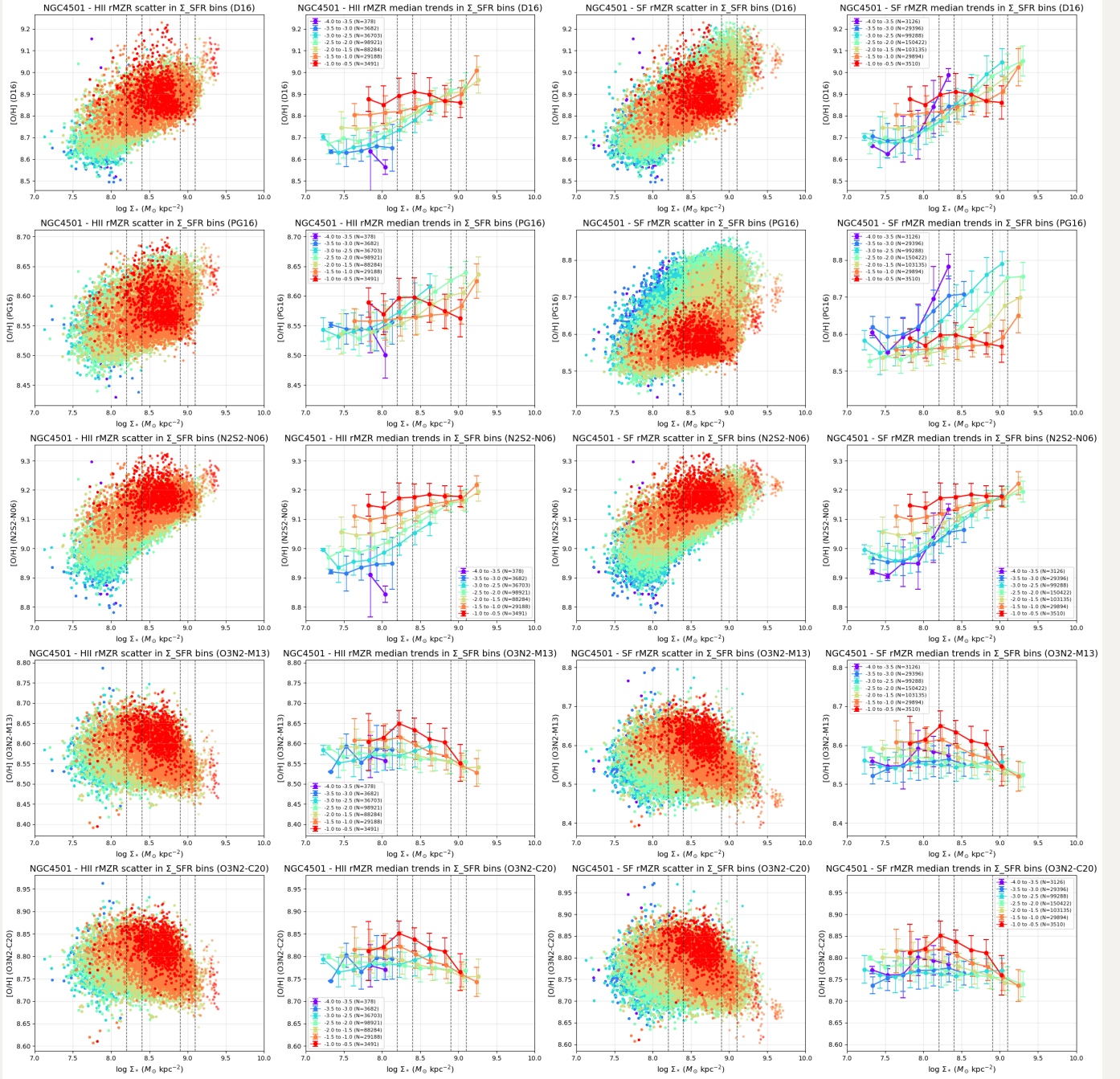
NGC4501

Total pixels: 1489380

Common valid HII pixels across all calibrations: 260851

Common valid SF pixels across all calibrations: 419018

NGC4501 - rMZR behavior in Σ_{SFR} bins: HII vs SF regions (Min 50 datapoints, Adjacent bins only)



Last week we have seen that when we exclude Comp region from SF region, we see the following 4 phenomena:

1. The abnormal deviation in Scal-PG16 (second row) in SF region (fourth column) become more consistent in HII region (second column), or at least behaves similarly to N2S2H α -D16 (first row) as expected.
2. Both O3N2 indicators (fourth and fifth rows) almost remain unchanged after removing Comp region. So they are less sensitive to accurate BPT classification (non-photonization)?
3. I add N2S2-N06 (third row) as well. So again, i am still concerned that the

increasing rMZR trends we see in first 3 rows is solely because we are tracing the Nitrogen abundance, not the actual metallicity. But we will get back to this question when we start to analyse rMZR.

4. All calibrations kind of show that higher metallicity at higher Σ_{SFR} .

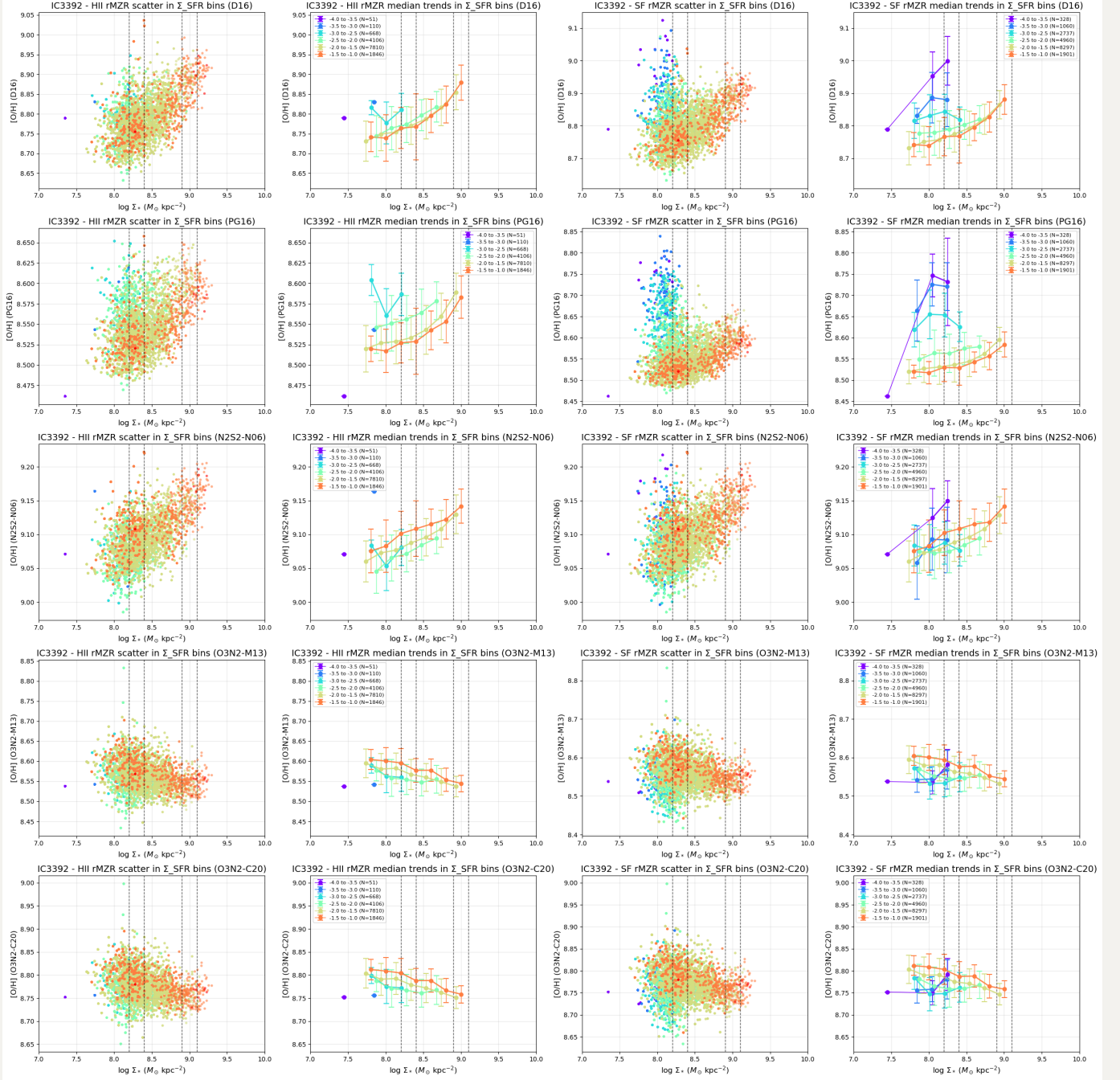
IC3392

Total pixels: 191406

Common valid HII pixels across all calibrations: 14884

Common valid SF pixels across all calibrations: 19776

IC3392 - rMZR behavior in Σ_{SFR} bins: HII vs SF regions (Min 50 datapoints, Adjacent bins only)



The first 3 statements are agreed, but it seems that N2S2H α -D16 and Scal-PG16 (first and second rows) shows lower metallicity at higher Σ_{SFR} .

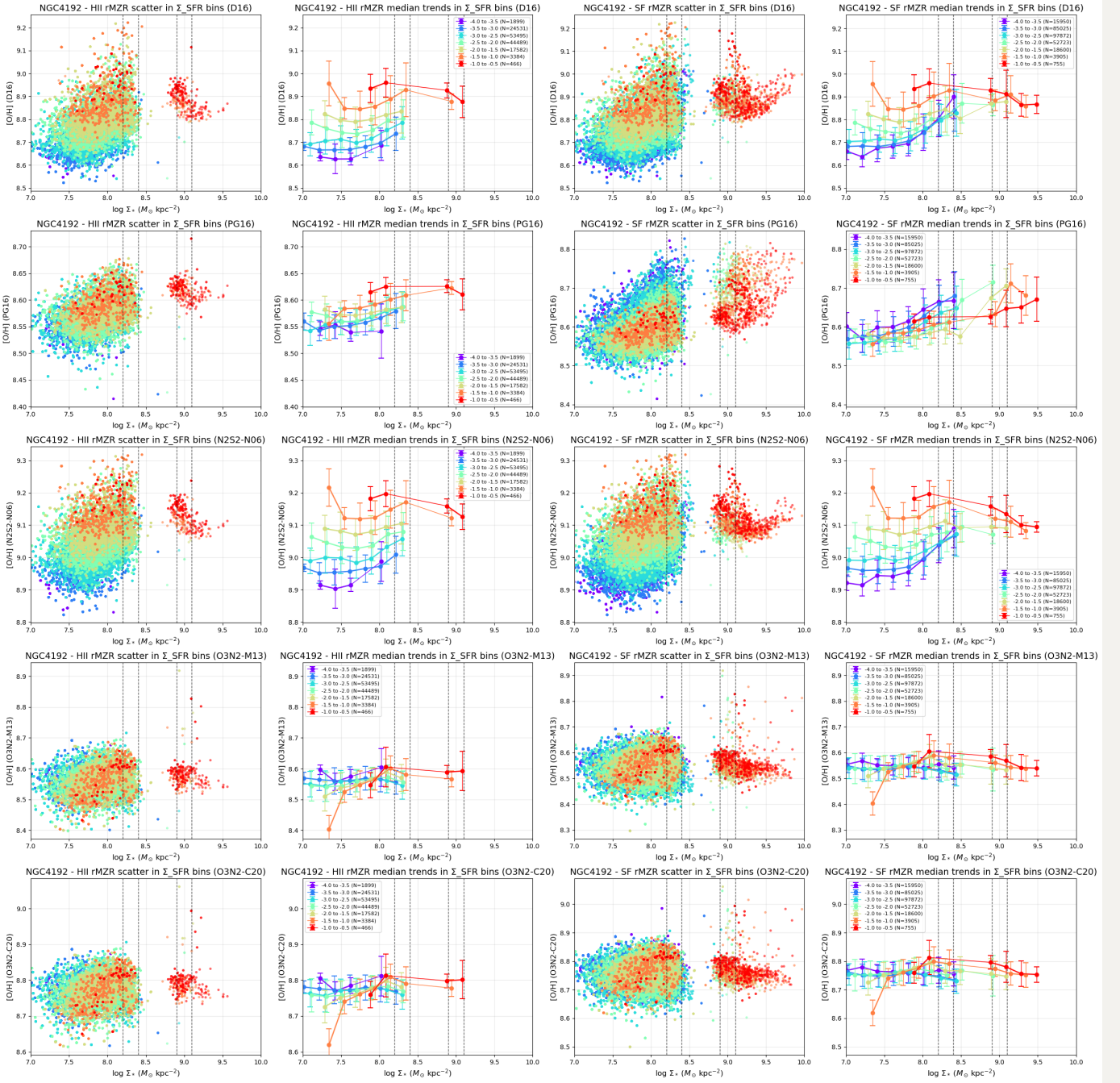
NGC4192

Total pixels: 1778868

Common valid HII pixels across all calibrations: 145934

Common valid SF pixels across all calibrations: 275008

NGC4192 - rMZR behavior in Σ_{SFR} bins: HII vs SF regions (Min 50 datapoints, Adjacent bins only)



The first 3 statements are agreed, while the fourth statement is agreed by first 3 rows but can't tell by last 2 rows.

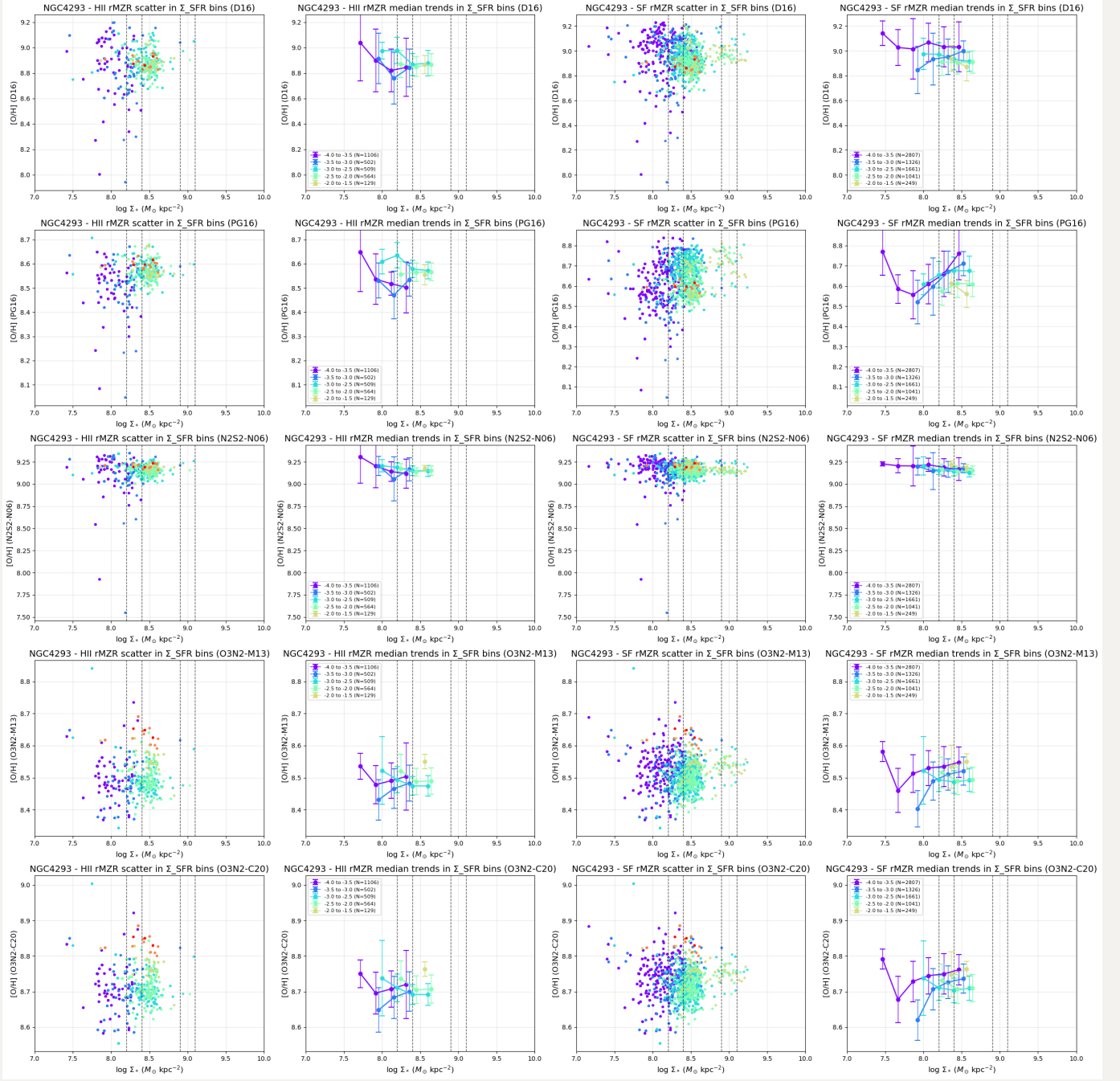
NGC4293

Total pixels: 491036

Common valid HII pixels across all calibrations: 3252

Common valid SF pixels across all calibrations: 7865

NGC4293 - rMZR behavior in Σ_{SFR} bins: HII vs SF regions (Min 50 datapoints, Adjacent bins only)



Too less spaxels so we may skip this galaxy.

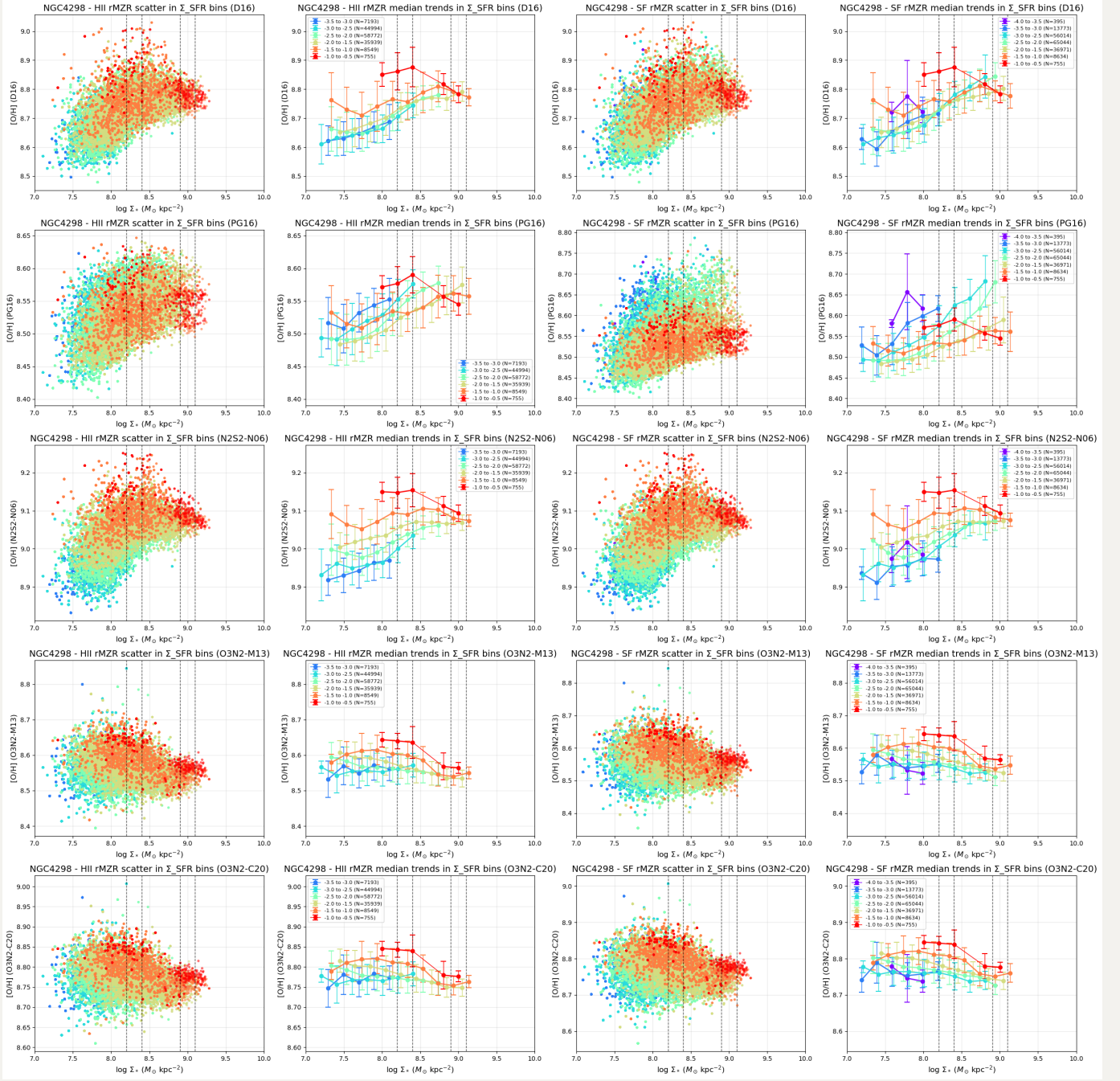
NGC4298

Total pixels: 1036790

Common valid HII pixels across all calibrations: 156350

Common valid SF pixels across all calibrations: 181696

NGC4298 - rMZR behavior in Σ_{SFR} bins: HII vs SF regions (Min 50 datapoints, Adjacent bins only)



All 4 are agreed.

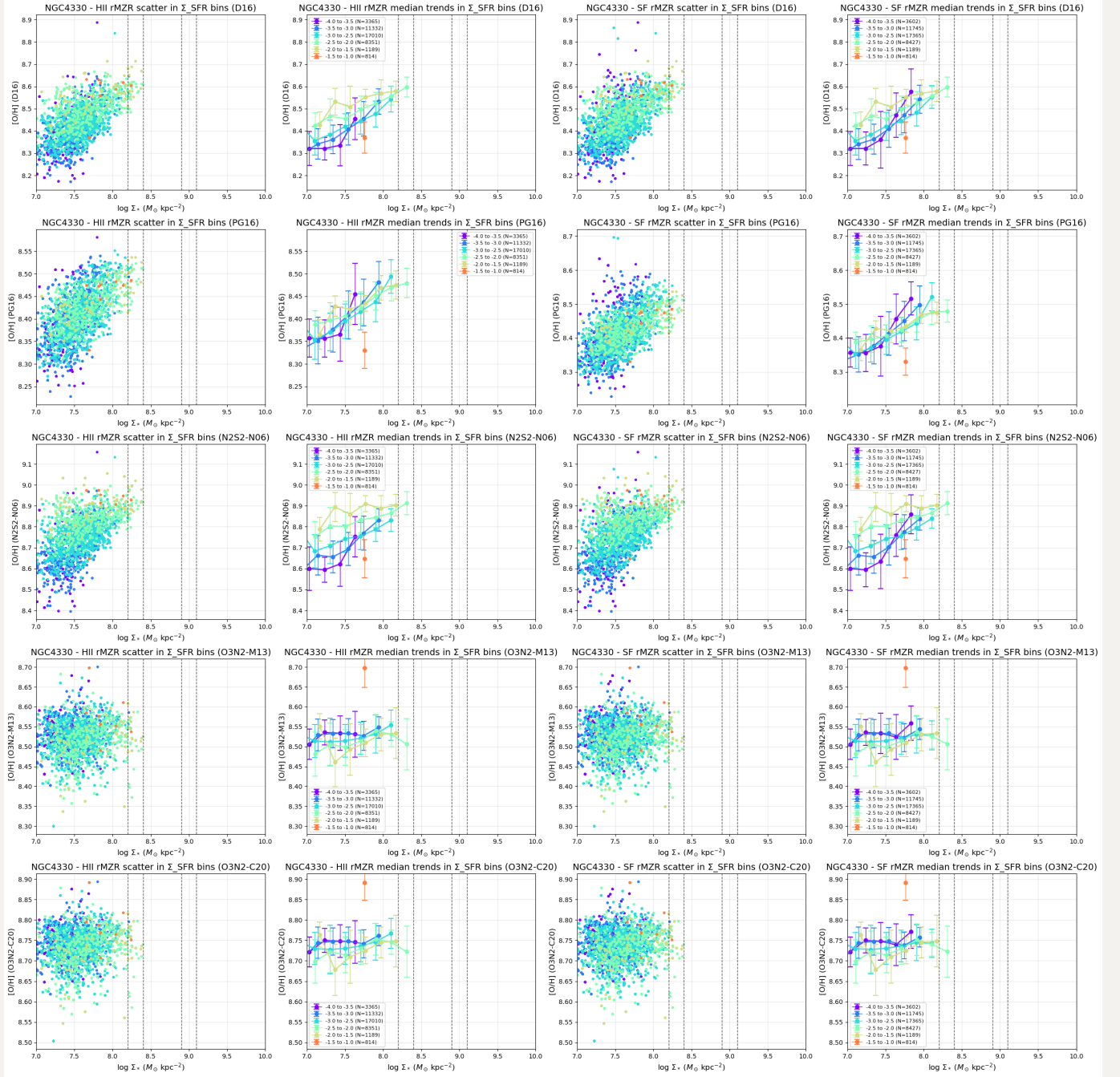
NGC4330

Total pixels: 669560

Common valid HII pixels across all calibrations: 42492

Common valid SF pixels across all calibrations: 43635

NGC4330 - rMZR behavior in Σ_{SFR} bins: HII vs SF regions (Min 50 datapoints, Adjacent bins only)



The first 3 statements are agreed, but it seems that both O3N2 (last 2 rows) shows lower metallicity at higher Σ_{SFR} .

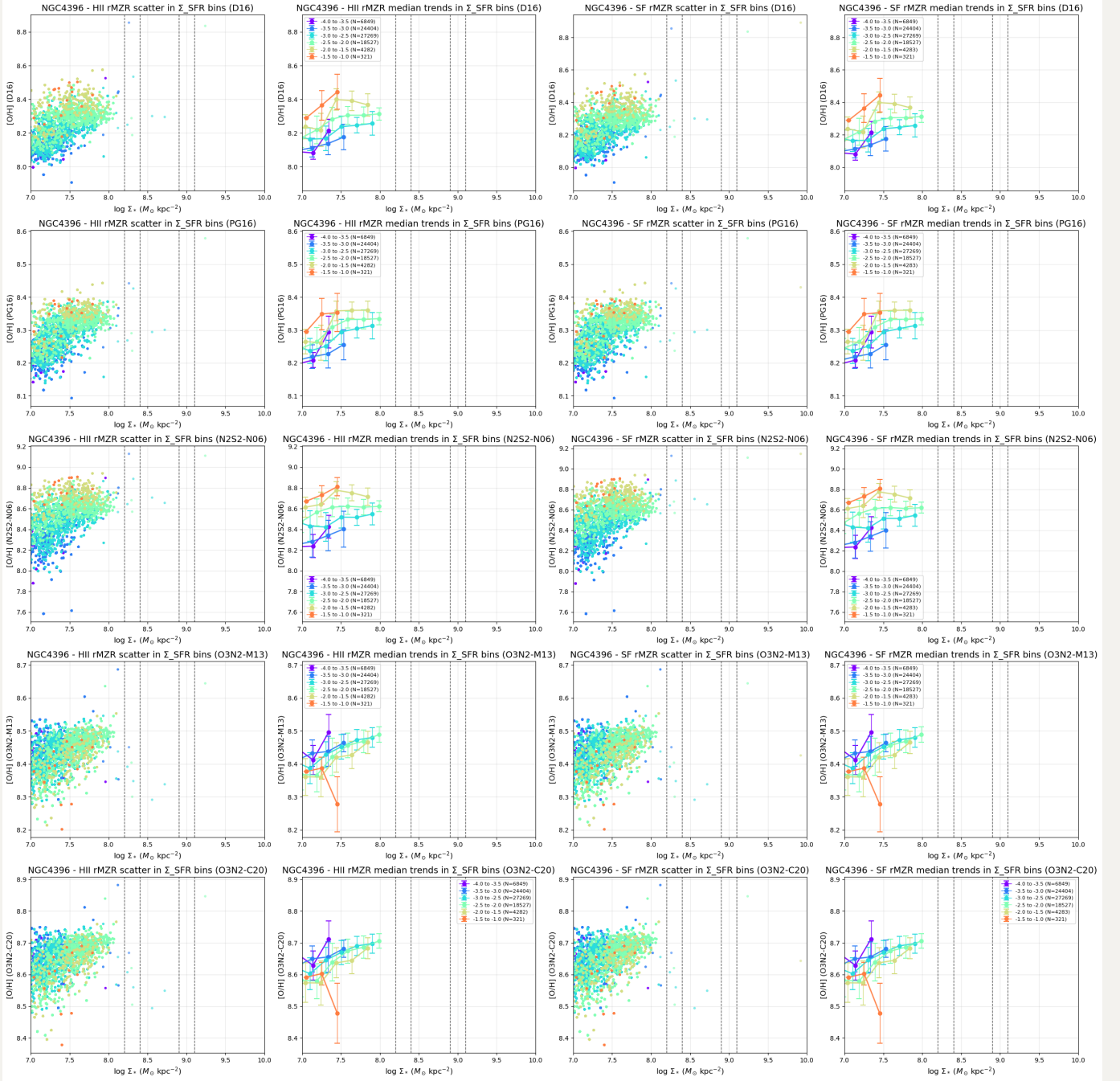
NGC4396

Total pixels: 666665

Common valid HII pixels across all calibrations: 81725

Common valid SF pixels across all calibrations: 81726

NGC4396 - rMZR behavior in Σ_{SFR} bins: HII vs SF regions (Min 50 datapoints, Adjacent bins only)



The first 3 statements are agreed, but it seems that N2S2H α -D16 and Scal-PG16 (first and second rows) shows lower metallicity at higher Σ_{SFR} .

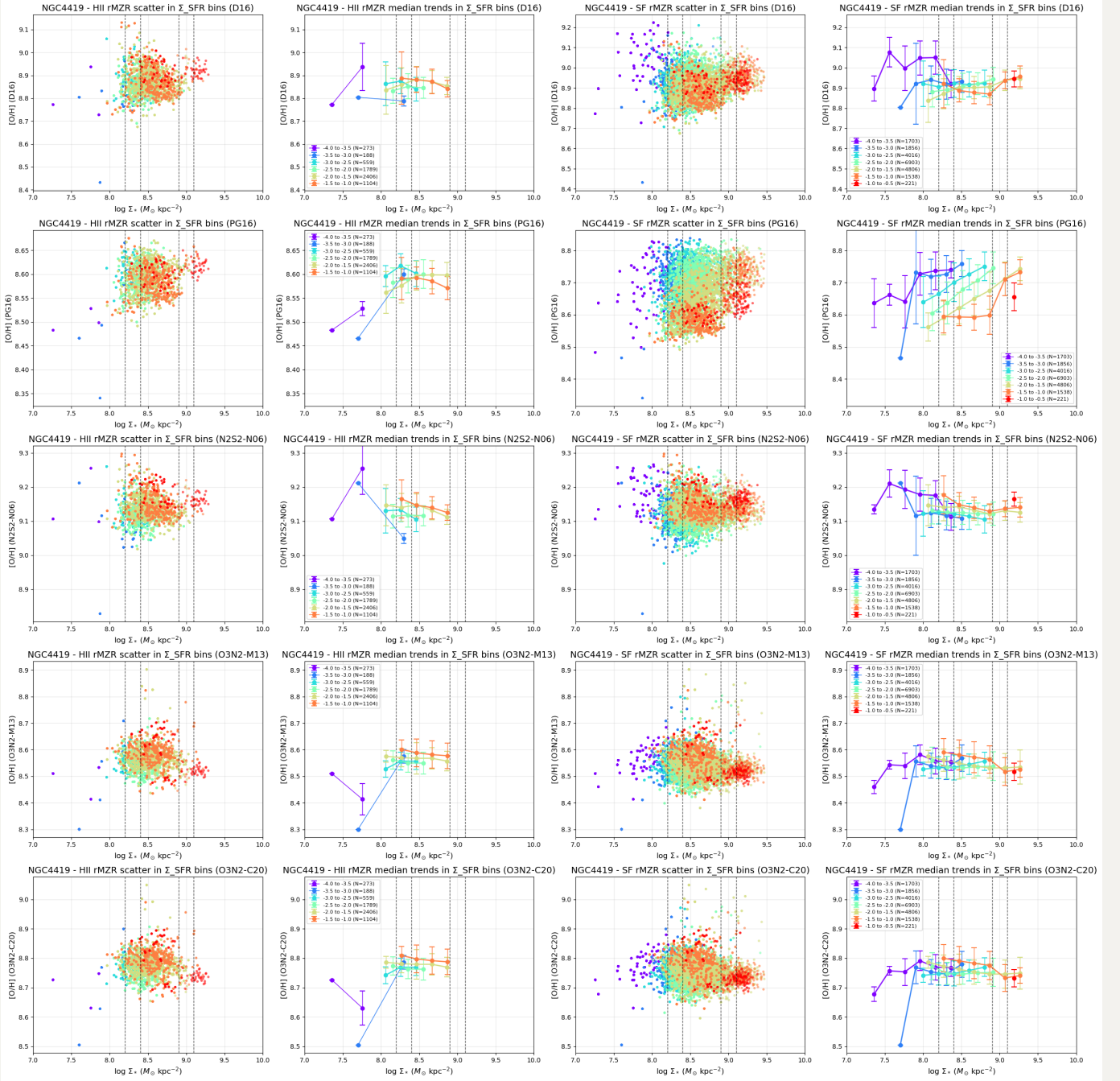
NGC4419

Total pixels: 403804

Common valid HII pixels across all calibrations: 6839

Common valid SF pixels across all calibrations: 21955

NGC4419 - rMZR behavior in Σ_{SFR} bins: HII vs SF regions (Min 50 datapoints, Adjacent bins only)



Too less spaxels so we may skip this galaxy, but seems all 4 are agreed except no clear increasing rMZR trends.

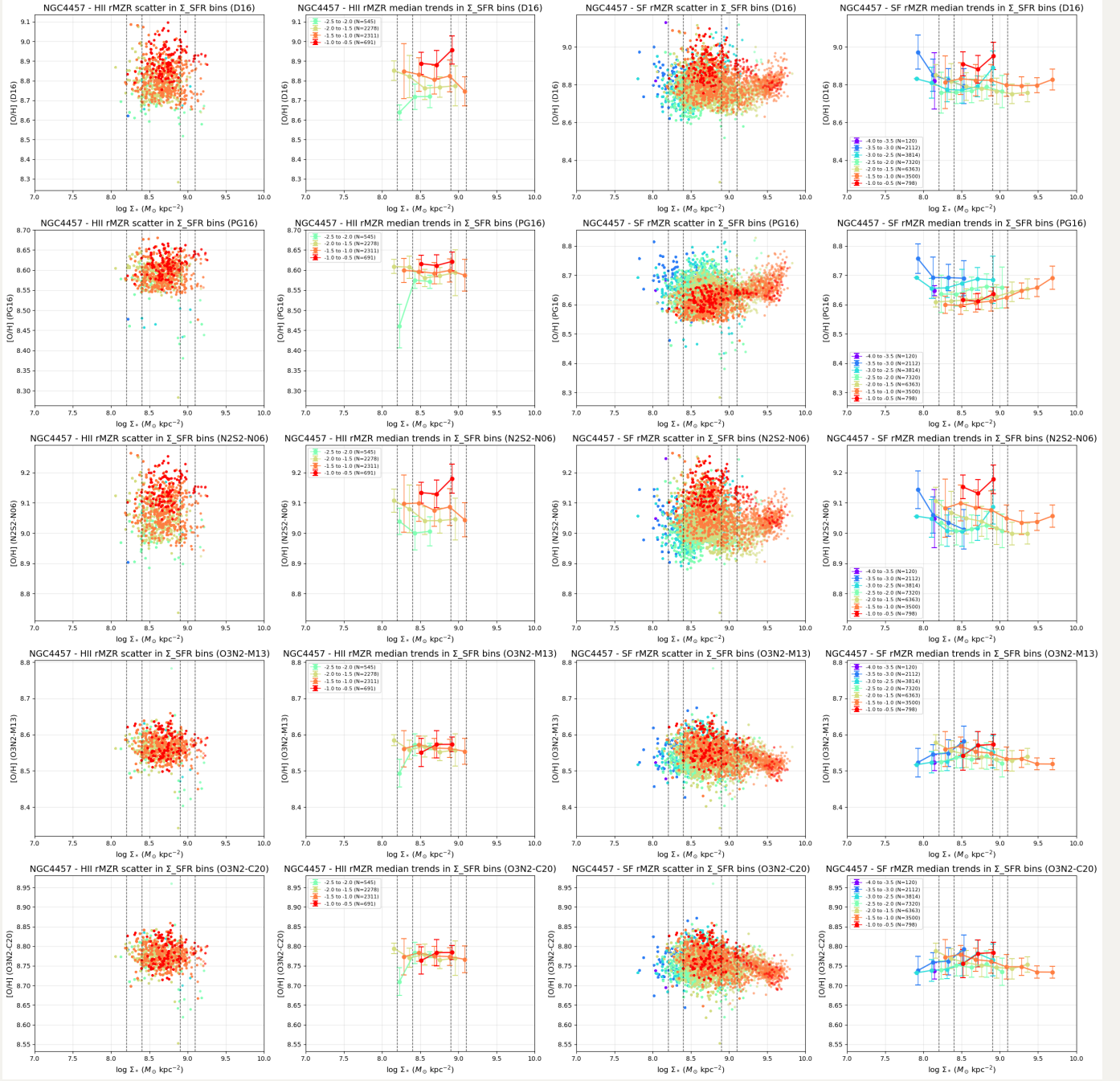
NGC4457

Total pixels: 525625

Common valid HII pixels across all calibrations: 5938

Common valid SF pixels across all calibrations: 24074

NGC4457 - rMZR behavior in Σ_{SFR} bins: HII vs SF regions (Min 50 datapoints, Adjacent bins only)



Too less spaxels so we may skip this galaxy.

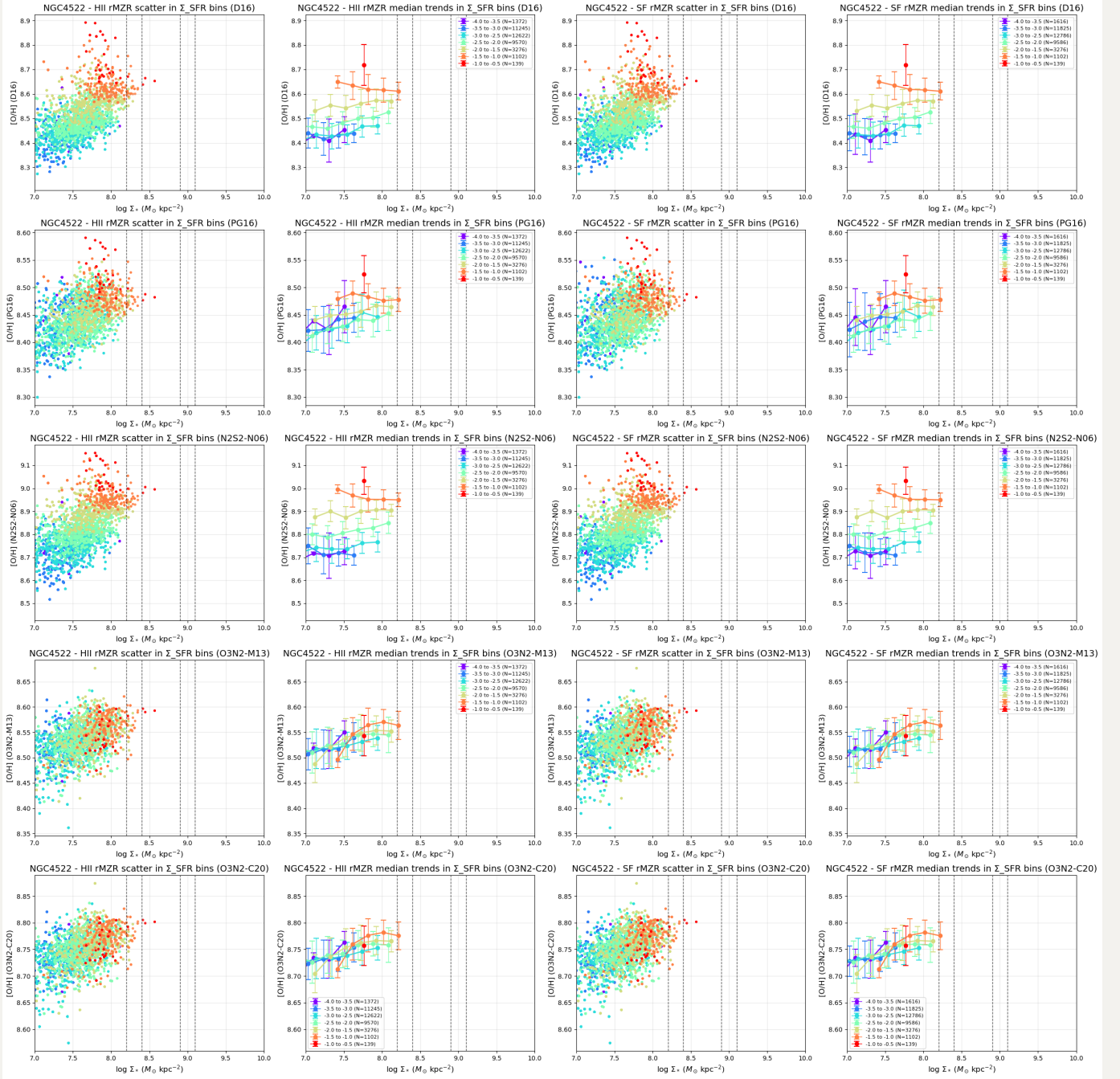
NGC4522

Total pixels: 388810

Common valid HII pixels across all calibrations: 39331

Common valid SF pixels across all calibrations: 40421

NGC4522 - rMZR behavior in Σ_{SFR} bins: HII vs SF regions (Min 50 datapoints, Adjacent bins only)



All 4 are agreed. Also notice that almost all the SF spaxels are in HII region.

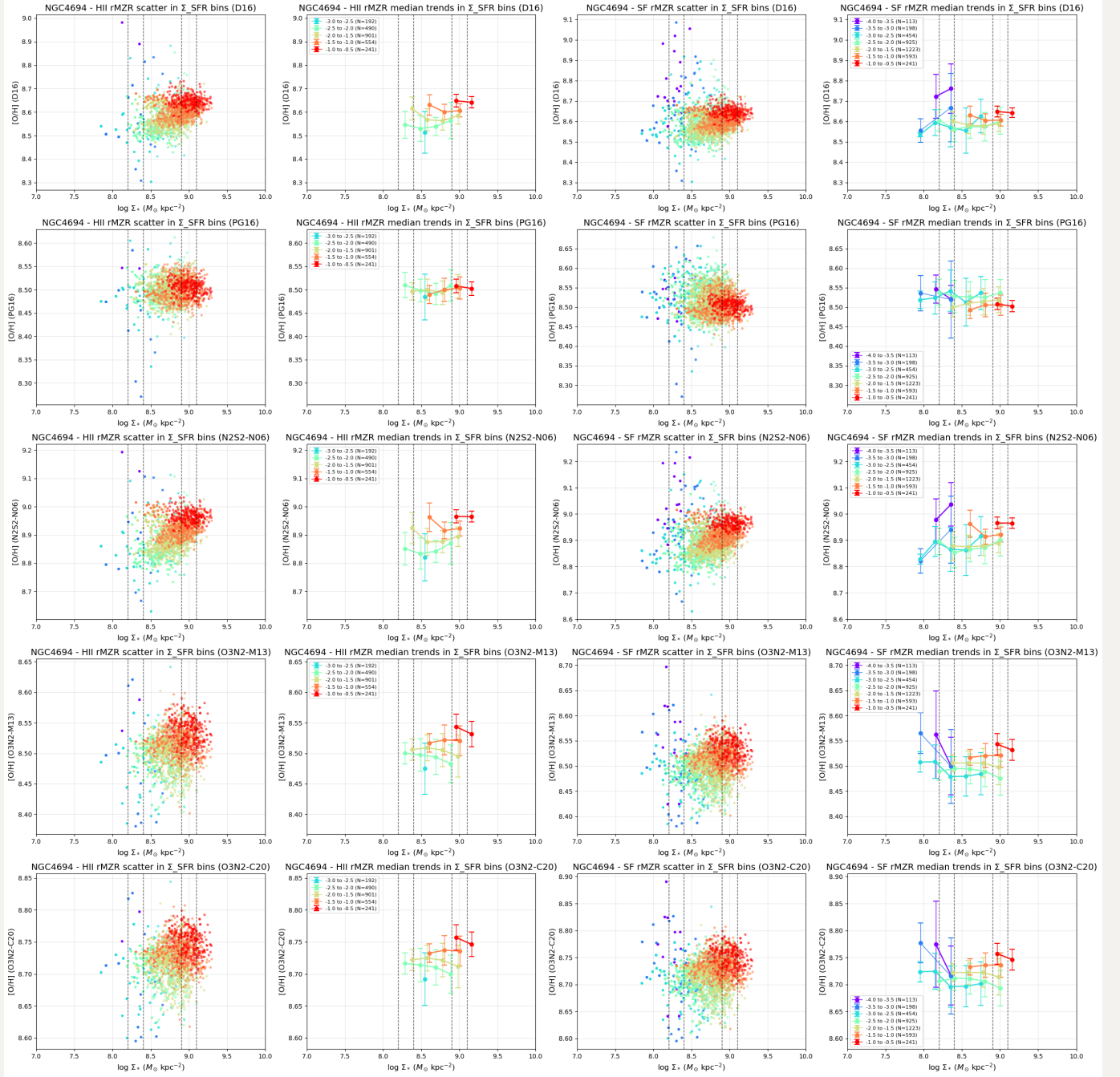
NGC4694

Total pixels: 193160

Common valid HII pixels across all calibrations: 2583

Common valid SF pixels across all calibrations: 3848

NGC4694 - rMZR behavior in Σ_{SFR} bins: HII vs SF regions (Min 50 datapoints, Adjacent bins only)



Too less spaxels so we may skip this galaxy, but seems all 4 are agreed except no clear increasing rMZR trends.

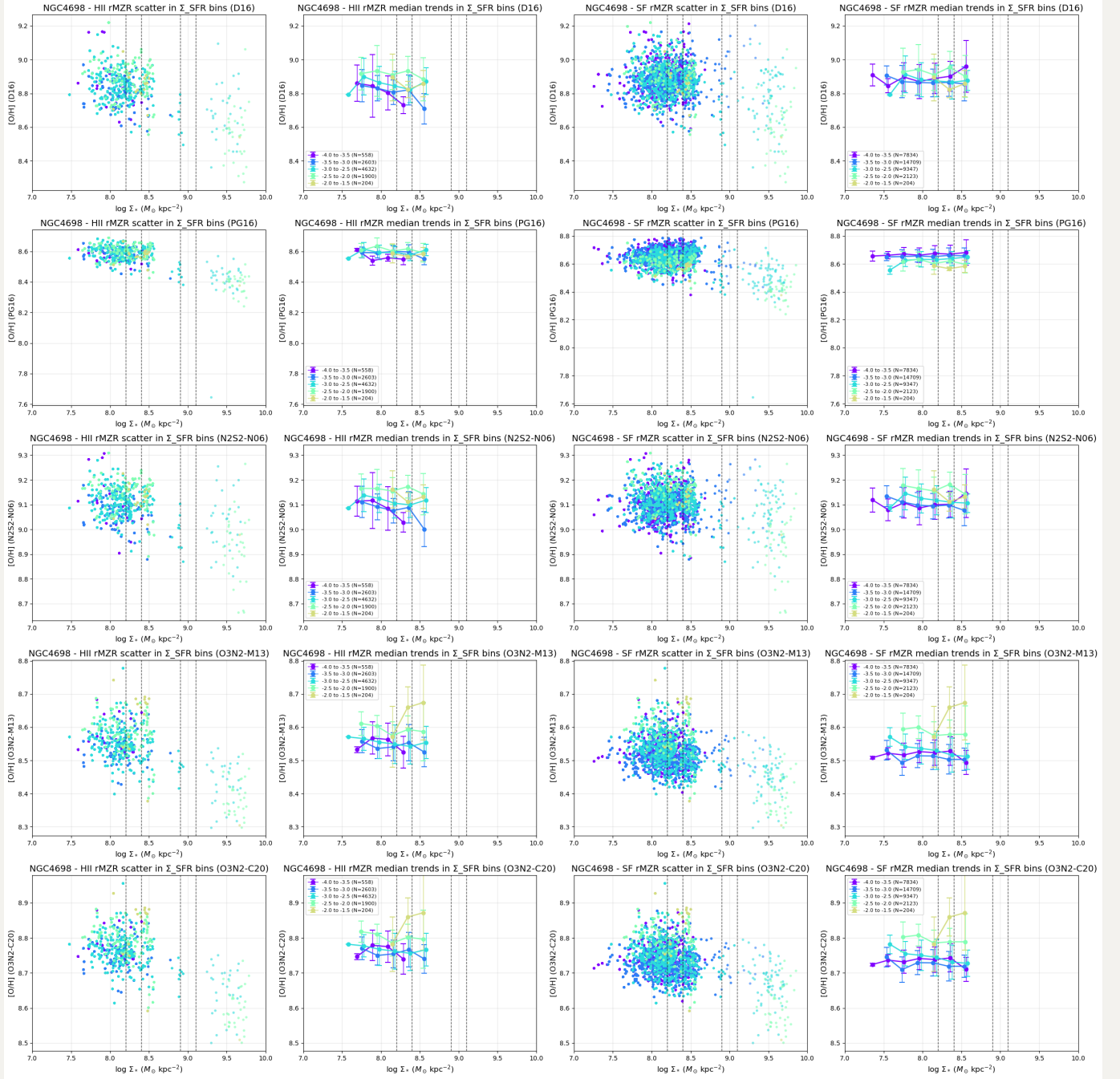
NGC4698

Total pixels: 722912

Common valid HII pixels across all calibrations: 10040

Common valid SF pixels across all calibrations: 34411

NGC4698 - rMZR behavior in Σ_{SFR} bins: HII vs SF regions (Min 50 datapoints, Adjacent bins only)



Too less spaxels so we may skip this galaxy, but seems all 4 are agreed except no clear increasing but maybe decreasing rMZR trends.

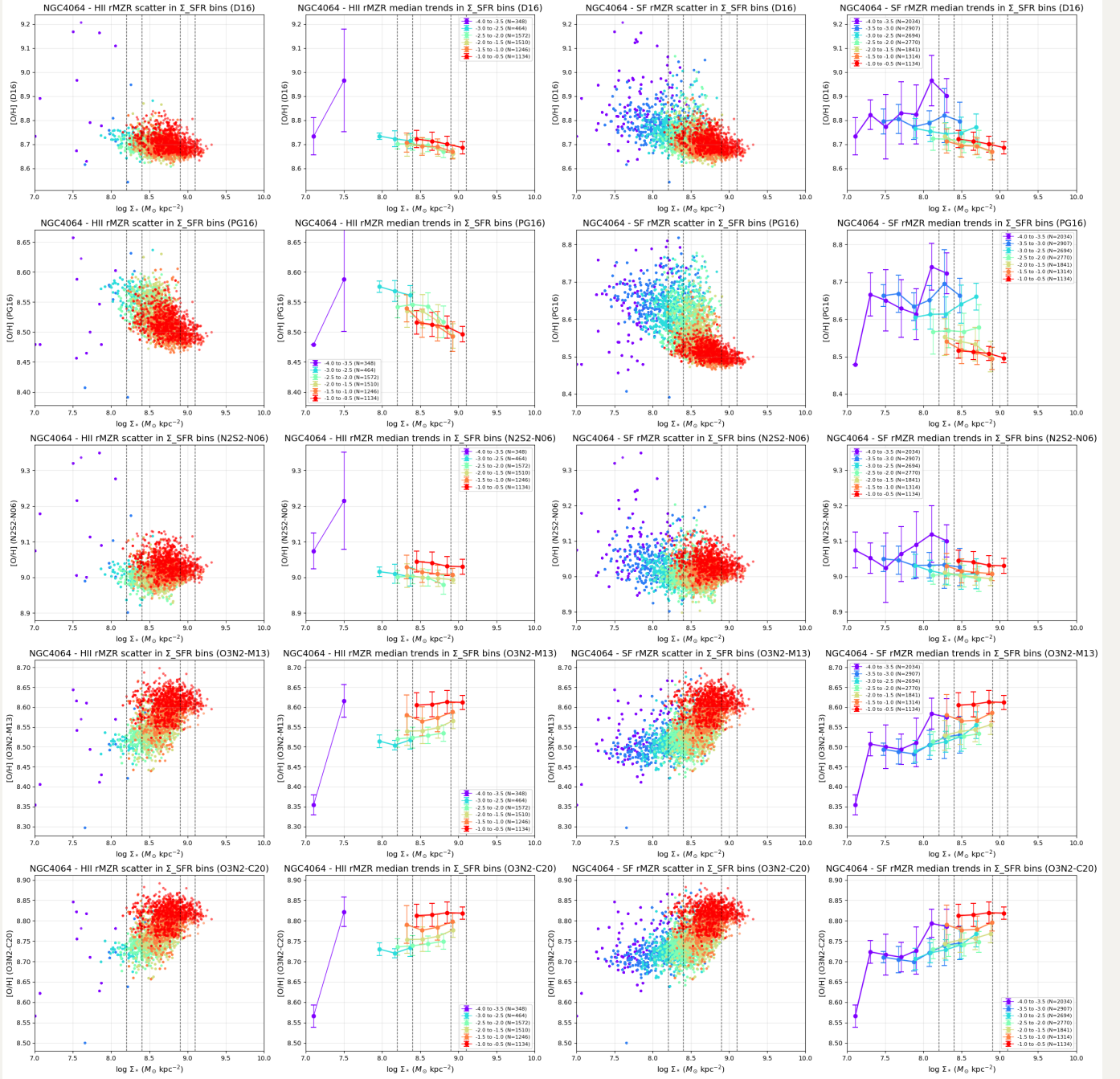
NGC4064 (outflow)

Total pixels: 376390

Common valid HII pixels across all calibrations: 6820

Common valid SF pixels across all calibrations: 15383

NGC4064 - rMZR behavior in Σ_{SFR} bins: HII vs SF regions (Min 50 datapoints, Adjacent bins only)



Too less spaxels so we may skip this galaxy, but the first 2 statements are agreed. It seems that N2S2H α -D16 and Scal-PG16 (first and second rows) shows lower metallicity at higher Σ_{SFR} , and decreasing rMZR trends.

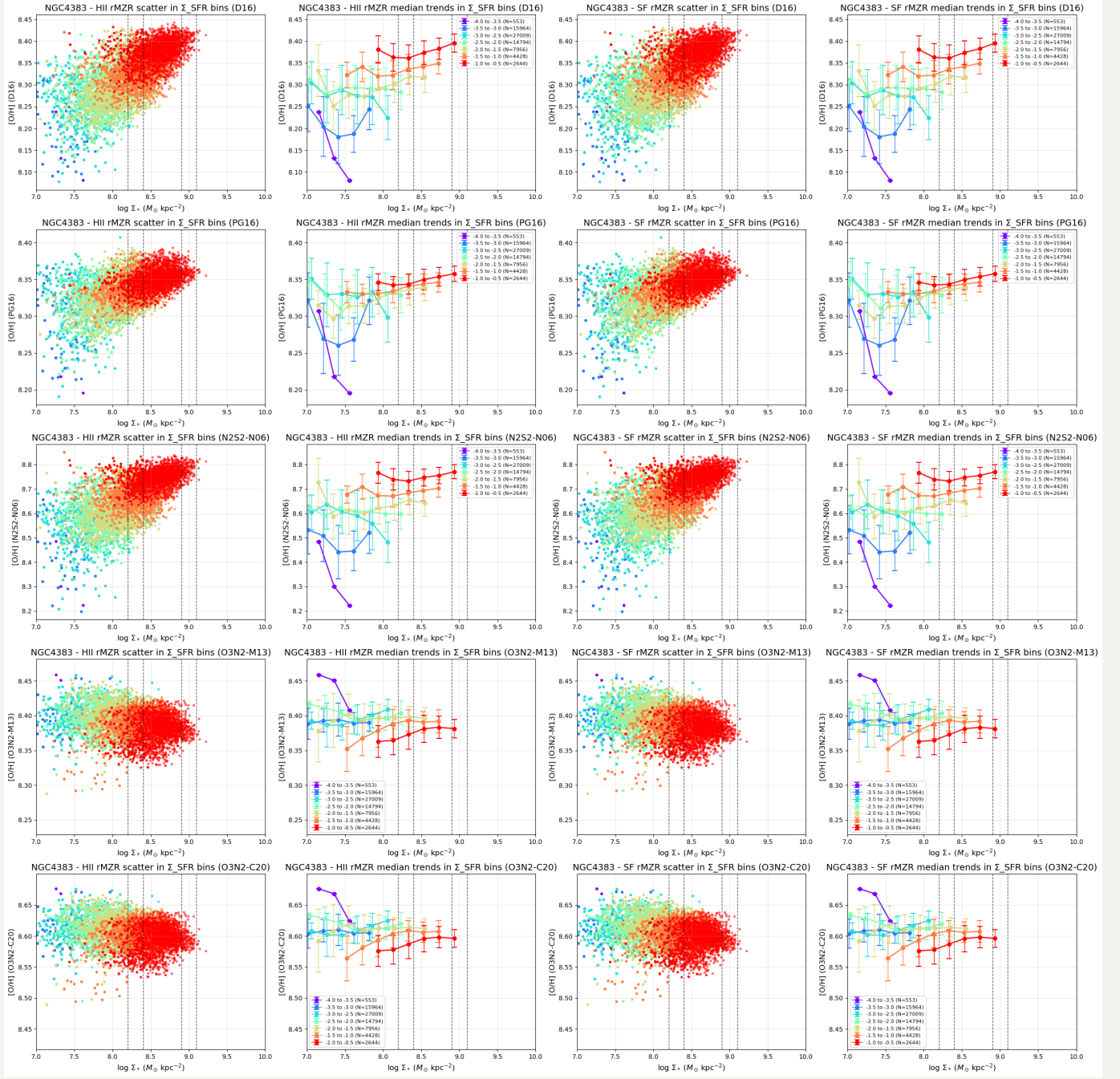
NGC4383 (outflow)

Total pixels: 446265

Common valid HII pixels across all calibrations: 75127

Common valid SF pixels across all calibrations: 75127

NGC4383 - rMZR behavior in Σ_{SFR} bins: HII vs SF regions (Min 50 datapoints, Adjacent bins only)



The first 3 statements are agreed, but it seems that both O3N2 (last 2 rows) shows lower metallicity at higher Σ_{SFR} .